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RESEARCH ARTICLE

Efficacious Novel Intrusion Detection System for Cloud Computing Environment

POOJA RANA¹, ISHA BATRA¹, ARUN MALIK¹, IN-HO RA², (Member, IEEE), OH-SUNG LEE³, AND A. S. M. SANWAR HOSEN⁴

¹School of Computer Science and Engineering, Lovely Professional University, Phagwara, Punjab 144401, India

²School of Computer, Information, and Communication Engineering, Kunsan National University, Gunsan 54150, South Korea

³School of Electronic and Information Engineering, Kunsan National University, Gunsan 54150, South Korea

⁴Department of Artificial Intelligence and Big Data, Woosong University, Daejeon 34606, South Korea

Corresponding authors: In-Ho Ra (ihra@kunsan.ac.kr) and A. S. M. Sanwar Hosen (sanwar@wsu.ac.kr)

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ABSTRACT Rife acceptance of Cloud Computing has made it bull's eye for the hackers. Intrusion detection System (IDS) plays a vibrant role for it. Researchers have done marvelous works on the development of a competence IDS. But there are many challenges still exists with IDS. One of the biggest concerns is that the computational complexity and false alarms of the IDS escalates with the increase in the number of features or attributes of the dataset. Hence, the concept of Feature Selection (FS) contributes an all-important role for the buildout of an efficacious IDS. New FS algorithm is put forward which is the modified Firefly Algorithm in which Decision Tree (DT) classifier is used as the classification function. We have used the hybrid classifier which is the combination of neural network and DT. We have used CSE CIC IDS 2018 dataset and simulated dataset for performance assessment. Our examination pragmatic that the performance of proposed architecture is better than the state-of-the-art algorithms.

INDEX TERMS Decision Tree, Firefly Algorithm, Fitness Function, Feature Selection, Genetic Algorithm, Intrusion Detection System, K-nearest Neighbor, Particle Swarm Optimization, Support Vector Machine, Neural Network, Random Forest.

I. INTRODUCTION

Cloud Computing (CC) acquires countless value today. It has on-demand and scalable services. It is satisfying the demand of its users by reducing overall cost and complexities [1]. Diverse types of cloud provide diverse services which fascinates sundry hackers. Intrusion Detection System (IDS) is very imperative. Due to huge traffic generation, cloud computing is becoming an eye-catching target for the attackers. The foremost security concern in this domain is to protect it from different network attacks. IDS ranges from anti-virus software to the well-developed monitoring system. Large data produced by the cloud is a biggest concern [2].

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The intrusions concoct the system capricious for the network traffic due to its nonlinear behavior. It is a proactive technology which monitors the malicious activities in the network and provides a protective mechanism. There is urgent need of providing good techniques for detecting attacks. It observes, collects and analyzes the network traffic, log files and actions of users for discovering the malicious activities in the network. Figure 1 represents different IDSs. On the basis of objective for the protection, there are two types of IDSs that are Host-Based IDS and Network-Based IDS where former is monitoring specific hosts and latter is monitoring network for the detection of malicious activities. On the basis of detection technique, there are two types of IDSs that are Anomaly-based and Signature-based where former is for the detection of attacks which are unknown or known and latter is used for known attack detection. IDS related to cloud can be applied to mobile e-health and resource-limited devices.

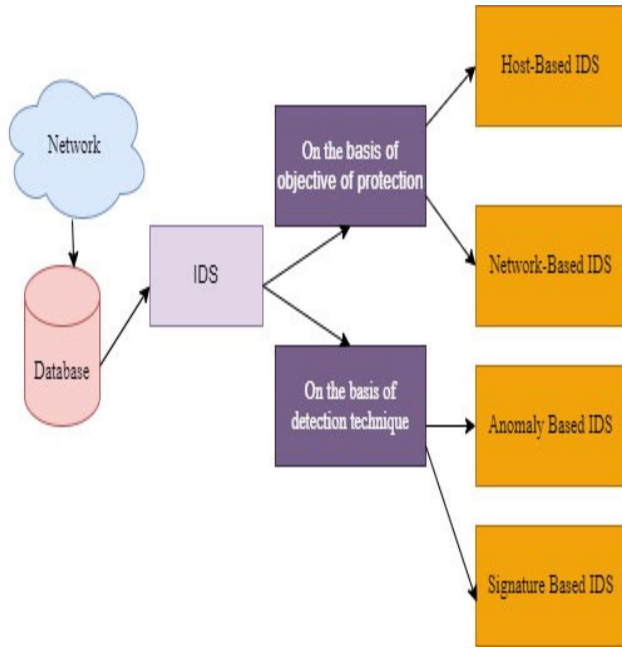


FIGURE 1. Different types of intrusion detection system (IDS).

Data mining and Machine Learning (ML) are used by various researchers [3]. Machine classifiers are usually used for binary classification into normal and attack packets [4].

Neural Networks (NNs) are widely used as they can perform very well on incomplete datasets [5]. While each service model in CC has its own benefits, they also encounter particular obstacles. In the case of Infrastructure as a Service (IaaS), virtualization is essential for infrastructure provisioning but has its limitations, and the value of IaaS services may decrease over time. Platform as a Service faces challenges related to interoperability, host sensitivity, confidentiality, authorization, reliability, and extensibility. On the other hand, Software as a Service deals with security concerns regarding authorization, authentication, data protection, reliability, and network monitoring. Cloud companies need to tackle these security challenges head-on. Feature Selection (FS) is a basic pre-processing step when the concept of Machine Learning (ML) comes. FS is important for an IDS as it enhances the performance and accuracy and it diminishes the number of features of the dataset or the network record [6], [7]. With the increase in the data accumulation in the databases and data warehouses, the dimensionality problem is becoming a big expostulation for the ML chores [8].

For making an efficacious IDS, there is an exigency of identification of the noteworthy features before the detection of attacks in the network. But the recognition of the significant features is a complicated task because the features can be relevant, extraneous or excessive which hikes the computation complexity for the detection of the attacks or intrusions [9], [10], [11]. Different techniques are used by researchers for developing good attack finding system in by using neural networks.

FS is aiming at retaining the relevant features which are required for building a strong model. For developing an IDS, it is vital to get rid of the irrelevant features so that the accuracy of the IDS increases. The highlights of our contributions are as follows:

- Discerned the various FS techniques and IDSs linked with the CC.
- Proposed a novel FS algorithm by modifying the Firefly Algorithm.
- Proposed a novel architecture for the detection of various attacks affecting CC.
- Latest intrusion detection dataset CSE-CIC-IDS 2018 and simulated dataset are used for the experiment and evaluation.

II. LITERATURE REVIEW

In [12], a model called Game Theory-based Cloud Security Deep Neural Network (GT-CSDNN) was proposed. This model incorporates both defender and attacker techniques using game theory principles. The main goal is to classify a network administrator to be aware of the nature of such traffic in order to effectively block and terminate any intrusive network connections. To achieve this, the Binary-Based Particle Swarm Optimization (BPSO) technique was utilized to identify the most relevant network features, while the Standard-Based Particle Swarm Optimization (SPSO) was used to fine-tune the control parameters of the Support Vector Machine (SVM).

In [17], authors evaluated and presented a detector based on Radial Basis Function Neural Network (RBF-NN) for detecting DDoS attacks. However, the resulting network structure can often be insufficient or unnecessarily complex, requiring manual configuration through a trial-and-error approach. This study proposes the use of the Bat Algorithm (BA) to automatically configure the RBF-NN network structure.

In [18], authors introduced a highly effective approach called the Dragonfly-Improved Invasive Weed Optimizer-based Shepard CNN (DIWO-based ShCNN) for detecting intruders and mitigating attacks in the cloud paradigm. This approach also enables the detection of intruders using ShCNN.

In another study [19], a powerful Intrusion Detection System (IDS) was proposed using the Sailfish Dolphin Optimizer-based Deep RNN (SFDO-based Deep RNN) to identify anomalies in the cloud framework. The SFDO algorithm combines the Sailfish Optimizer (SFO) with the Dolphin Echolocation (DE) technique. The ChicWhale technique can be utilized for Virtual Machine (VM) migration and cloud data management.

In [20] conducted a study on an IDS (Intrusion Detection System) using a Fisher Kernel-Based PCA dimensional reduction technique and a Grey Wolf Optimizer (GWO)-based weight dropped Bi-LSTM technique (FKPCA-GWO WDBiLSTM). Firstly, they combined the data record with PCA to achieve linearly separable dimensionality reduction by using the fisher kernel with fisher score as input. Secondly,

they employed the WDBiLSTM network to retain long-term dependencies while eliminating features from both forward and backward directions.

In another study [21], a novel Deep Learning (DL) approach incorporating CNNs Convolutional Networks and Recurrent Neural Networks was developed for cloud security in IDS. With this DL technique, technique, they were able to prevent some detected but unauthorized traffic from accessing the server in the cloud.

The CC ecosystem includes three primary service models: IaaS, PaaS, and SaaS. These models are the essential building blocks of CC and can be deployed in different setups, such as community, private, hybrid, and public clouds [22]. Each service model offers distinct features to meet the diverse requirements of users.

IaaS, which forms the base level, furnishes virtualization, servers, storage, and network resources. It provides users with a versatile and expandable infrastructure for constructing and overseeing their applications [23]. PaaS builds on top of IaaS by presenting technical layers and instances of management software, empowering developers to concentrate on application development without concerns about the underlying infrastructure. Conversely, SaaS delivers fully functional software applications accessible through the cloud, enabling users to run applications without requiring local installations.

The ever-changing threat landscape prompts individuals to continuously adapt their methods in order to exploit vulnerabilities in cloud environments [24]. Traditional IDSs often require assistance in effectively identifying changes in network traffic patterns. In response, experts stress the significance of incorporating ML and DL techniques to enhance the capabilities of IDSs [25]. They have become increasingly important in various sectors, such as finance, government, scientific research, and security [26], [27]. ML's ability to cluster and classify data efficiently plays a crucial role in cybersecurity applications [28], [29].

An anomaly-based IDS examines real-time network traffic by comparing it to previously recorded patterns of normal behavior in order to identify new intrusions. While this approach is effective in detecting novel attacks, it can also generate false-positive alerts by mistakenly flagging regular packets as malicious [30], [31]. On the other hand, a misuse-based IDS relies on a signature database to detect known attacks, which helps reduce the occurrence of false alarms. However, this type of IDS may overlook new threats that do not have recognized signatures.

Random Forest (RF), a method of ensemble learning, is based on Decision Trees (DTs). During training, it constructs multiple DTs and produces the class, which can be the mode of the categories or the average prediction of the individual trees [32]. RF is particularly well-suited for our IDS model because it effectively handles high-dimensional datasets with numerous features. A Radial Basis Function Neural Network (RBFNN) is a type of Neural Network

that utilizes radial basis functions as activation functions in its hidden layer functions convert the input data into the hidden layer [33]. RBFNNs are highly effective when dealing with nonlinear problems and are especially suitable for tasks involving pattern recognition. They are well-suited for intrusion detection scenarios where attacks can be complex and nonlinear [33], [34]. The architecture of an RBFNN allows it to efficiently approximate complex decision boundaries, resulting in improved performance in capturing the underlying patterns in the data. Moreover, compared to traditional feedforward neural networks, the training process of an RBFNN is relatively faster, making it computationally efficient for large-scale intrusion detection tasks [33].

From literature review, following research gaps are observed:

1. Less work is done by researchers on the exploration of FFA.
2. Most researchers have used old datasets for the validation of their proposed work.
3. Most researchers have used benchmark dataset.
4. Hybridization in FS and classification module is found rarely in research work.

Following points show the contribution of proposed work:

1. FFA is used in modified form for
2. FS.
3. Latest attacks are detected by proposed work. Latest dataset is used for validation of dataset.
4. Simulated dataset is created for the checking the performance of the proposed work.
5. Hybridization is done in FS module and classification module for the development of an efficient IDS.

III. PROPOSED METHODOLOGY

This section describes the proposed architecture for various attacks detection in CC environment. The framework consists of three modules that are:

A. DATASET

This section of the paper is describing the datasets used for the performance assessment of the proposed algorithm. The objective of our proposed work is to assess the proposed work on recent dataset instead of most commonly old datasets like KDDCup 1999, NSL-KDD dataset. We have used CSE CIC IDS 2018 dataset [35] which is Intrusion Detection Evaluation Dataset which was covered in 5 days. This dataset is having around 2 lakh records with 80 features.

In [36], it is stated that real-time cloud infrastructures are very expensive and complex. Also, it is very difficult to re-configure the cloud parameters on a large scale. Using the cloudsim toolkit helps to recreate the outputs in an easy and controlled way. We have created our own dataset which is created by using the simulated dataset. We have used the Eclipse IDE for Java Developers - 2023-09 for the creation of dataset and used the cloudsim 3.0.3 jar files.

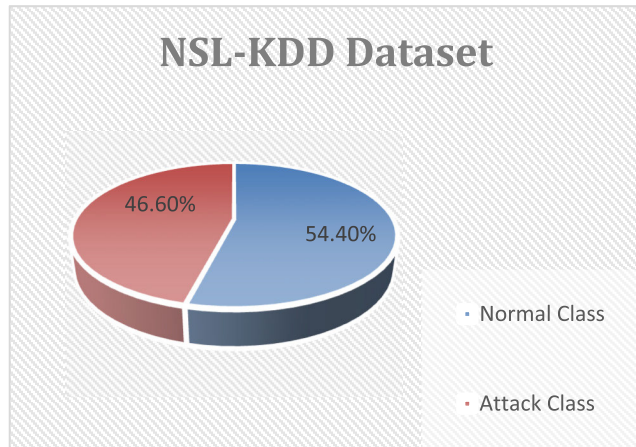


FIGURE 2. NSL-KDD dataset distribution.

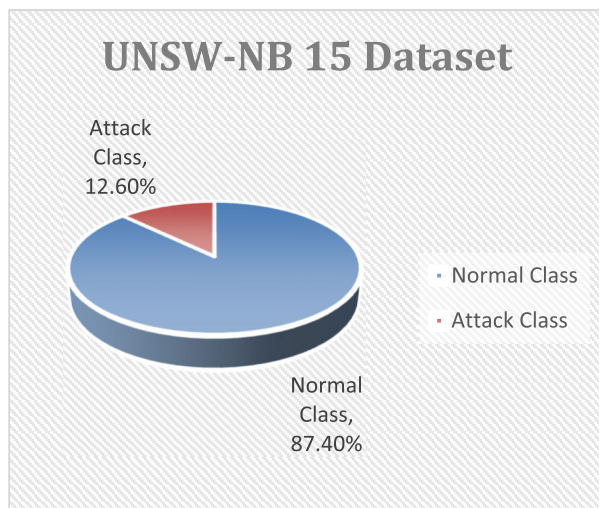


FIGURE 3. UNSW-NB 15 dataset distribution.

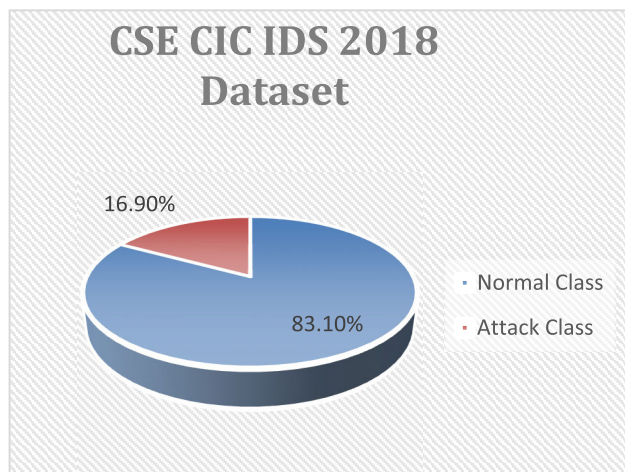


FIGURE 4. CSE CIC IDS 2018 dataset distribution.

B. PREPROCESSING MODULE

This section of the paper describes the function of pre-processing module. In lay man language, it cleans the data. The raw dataset has categorical and numerical data. The data can be irregular or skewed which may lead to find wrong results.

TABLE 1. Proposed feature selection algorithm.

Algorithm: Proposed FS Algorithm	
	Input: Dataset
	Initialization of the search space with n number of fireflies, dimension of search space is D and number of iterations is I. Initialization of Parameters. Parameters: n=Population Size (Number of fireflies) max_Generations (Maximum number of generations) alpha (Randomness factor) beta_min and beta_max (Minimum and Maximum Attraction Coefficients) gamma (Light Absorption Coefficient)
Step 2:	Initialize the population of fireflies $x_i=(i-1,2,...,n)$ with random positions.
Step 3:	Evaluate the objective function $f(x_i)$ for each firefly x_i .
Step 4:	for t=1 to max_generations do; for i=1 to n do; for j=1 to n do; if $f(x_i) < f(x_j)$ then: Calculate the distance r_{ij} between firefly x_i and x_j Calculate the attractiveness beta using: Update the position of firefly x_i ; End if End for End for Evaluate the objective function $f(x_i)$ for each firefly x_i End for
	Output: Optimized Feature Subset of the Dataset.

Machine learning techniques only works on numerical data which is cleaned. This module removes the null values from the dataset. Strings are converted to numeral form which is easily understood by the computer operating system.

Normalization is performed for putting all feature values into a specific range. It is performed for scaling dataset. It makes the values of the dataset in $[-1, 1]$ interval. We have used popular normalization technique Min-Max Normalization. A linear transformation is done by the Min-Max Normalization, refer to Eq. (1).

$$A' = \frac{(A - \text{min value of } A)}{(\text{max value of } A - \text{min value of } A)} * (D - C) + C \quad (1)$$

where A is the having the Min-Max Normalized Data A if predefined boundary belongs to $[C, D]$.

C. FEATURE SELECTION MODULE

FS performs an important protagonist in the classification. We have presented a modified Firefly Algorithm (FFA) for the FS which is the modification in the original FFA. The proposed FS algorithm has novelty as we have used the decision tree (DT) classifier as the fitness function for the FFA. FFA is chosen as it is very rarely used by researchers for FS. Literature survey has proved that FFA is better than Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) [37]. Some variants of PSO are the special cases of FFA. Parameters of FFA can be adjusted which can improve convergence.

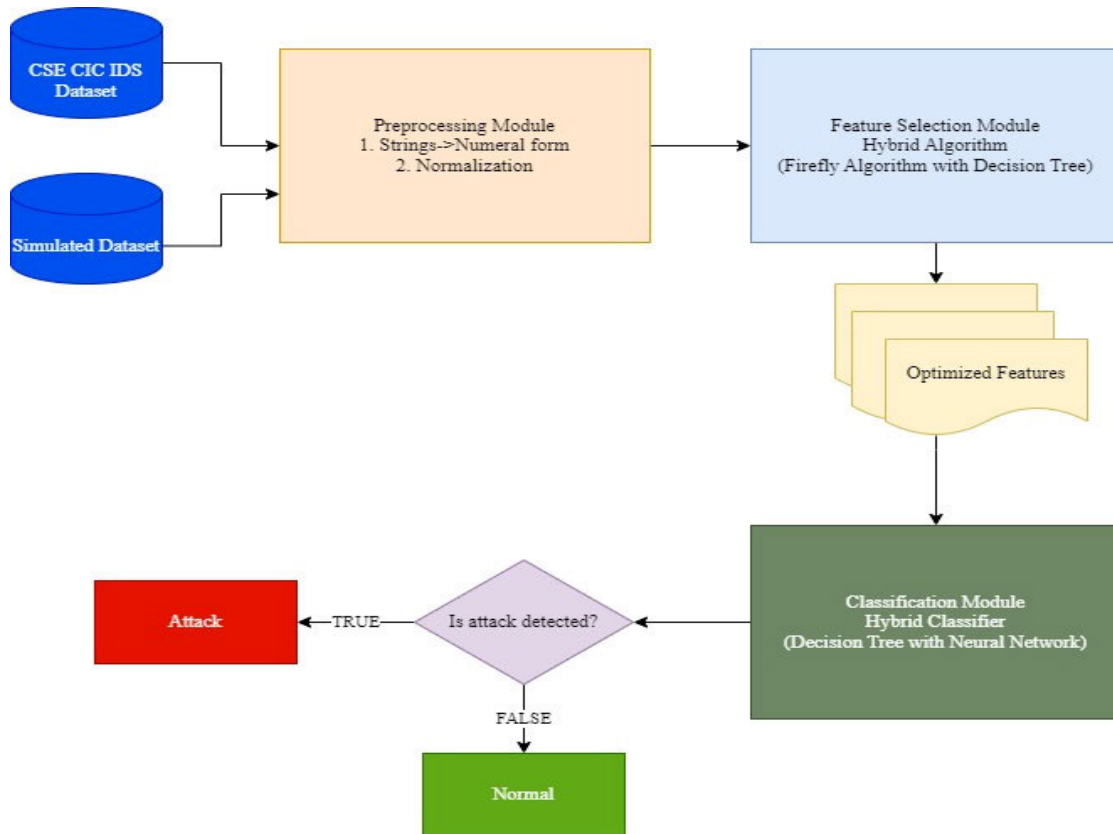


FIGURE 5. Workflow diagram of the proposed method.

FFA is presented by Xin-She Yang in 2007 which mimics the behavior of the fireflies in the summer in the tropical region. Fireflies search for food, communicate with each other and search mates by using the bioluminescence with different patterns of flashlight. FFA is based on the flashing features of the fireflies which generates a firefly-based meta-heuristic algorithm. The basic three principles of fireflies are:

- 1) Each and every firefly is having unisex. Hence, the attraction for mating is independent of the sex of the fireflies.
- 2) The power of attraction is proportional to the brightness of the fireflies. The firefly with less brightness will move towards the firefly with more brightness. Attraction between fireflies is inversely proportional to the distance between the fireflies as more distance less attraction.
- 3) The cost function in the analytical form is linked with the brightness of the firefly.

There is a question that why FFA is efficient algorithm. FFA is a swarm-based algorithm so it has same advantages as other swarm-based algorithms has. Some variants of PSO are representing FFA as accelerated PSO. FFA can be adapted to control the randomness and iterations proceed. All these above advantages of FFA makes to handle clustering problems, classification problems and continuous problems.

D. CLASSIFICATION MODULE

This section of the paper describes the classification module which is used for the detection of the attacks in the CC environment. Hybrid classifier is used for the classification of the attacks. We used the hybridization of the NN with the DT for the detection of the attacks in the CC environment.

NNs serve as a metaphor for how the brain processes information, drawing inspiration from biology rather than replicating its exact functions. They've proven highly promising in various forecasting and business classification tasks by leveraging their capacity to learn from data. This article seeks to offer a concise introduction to NN. These networks learn by adjusting their architecture and connection strengths, enabling them to adeptly handle tasks. They acquire knowledge either from provided training data or by autonomously grasping patterns from input-output relationships and examples.

DTs offer a widely embraced and efficient method for classification. They take the form of a flowchart-like structure, employing a tree model to visualize decisions and their potential outcomes. Constructed from existing training data, each internal node conducts a test on a particular attribute, with each branch indicating the test outcome, while every leaf node signifies a class label or a definitive decision.

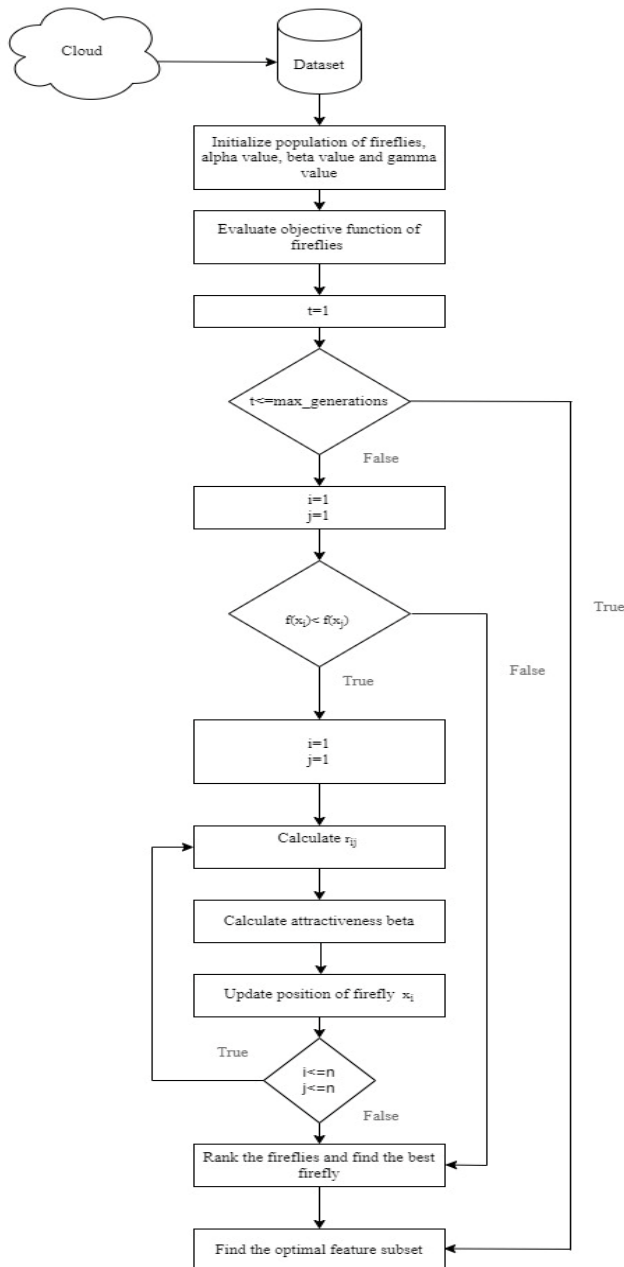


FIGURE 6. Flowchart diagram of Feature Selection which is combination of FFA and DT.

In the proposed work, the hybrid classification model comprising NN and DT components offers a promising approach to enhance the overall classification accuracy. By effectively combining the strengths of both models, it can overcome the limitations of individual algorithms and provide more accurate and interpretable predictions for the given classification task.

1. **Improved Robustness to Overfitting:** NNs, especially deep architectures, have a tendency to overfit the training data, particularly when dealing with small datasets or noisy input. DTs, on the other hand, are less prone to overfitting due to their inherent simplicity

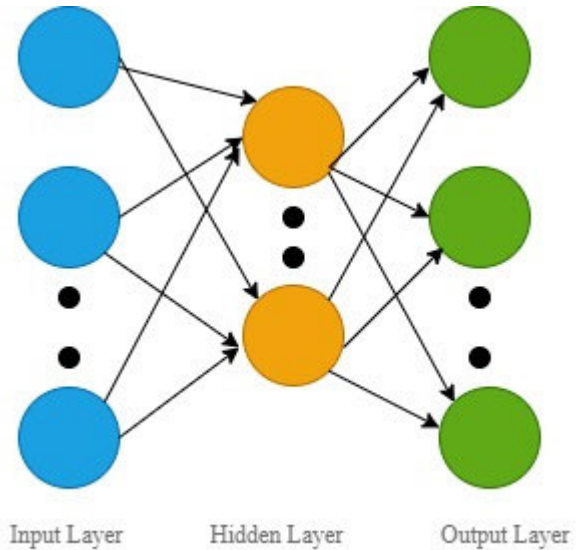


FIGURE 7. Representation of neural network.

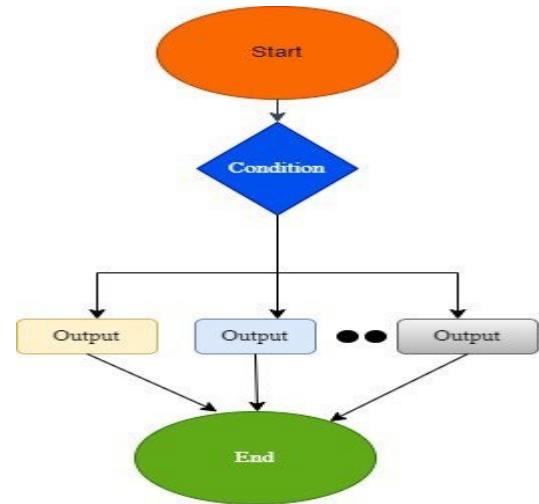


FIGURE 8. Representation of decision tree.

and ability to capture local patterns. By integrating DTs into the hybrid model, it can help mitigate the risk of overfitting, leading to more generalized and reliable predictions.

2. **Enhanced Feature Representation Learning:** NNs excel at automatically learning feature representations from raw data, extracting hierarchical and abstract features through successive layers of neurons. By pre-processing the data using NN layers before feeding it into the DT classifier, the hybrid model can provide better feature representations, potentially leading to improved discrimination between classes and higher classification accuracy.
3. **Flexible Model Architecture:** The hybrid model offers flexibility in designing the architecture and configuration of both NN and DT components. Researchers and practitioners have the freedom to experiment with

TABLE 2. Following table shows the precision comparison.

Total number of Samples	Precision Proposed Firefly + Hybrid Classifier	Precision Levenberg Neural	Precision DT	Precision RF	Precision KNN with 10 Neighbors	Precision Multi-SVM
20000	0.95885899	0.92957402	0.9275468	0.92419316	0.94224824	0.93979744
30000	0.95917421	0.94449384	0.94443984	0.94042102	0.93438557	0.93592328
40000	0.96476389	0.93675	0.93637754	0.93015736	0.93765343	0.93428337
50000	0.96498339	0.94046049	0.93662454	0.93755287	0.94267411	0.93886657
60000	0.96272396	0.93056105	0.93913262	0.94535597	0.93448569	0.94106818
70000	0.96415675	0.94553352	0.94136275	0.93481326	0.94053765	0.93824546
80000	0.95938611	0.9421243	0.93947408	0.94333487	0.94941953	0.93502283
90000	0.9586237	0.93736232	0.93514004	0.94924117	0.9346094	0.94853601
100000	0.95963389	0.94947545	0.94199408	0.93958422	0.94926324	0.94560917
110000	0.96626869	0.93482131	0.94661629	0.93885752	0.94579961	0.94611457
120000	0.95983295	0.93395744	0.93762854	0.94185481	0.94869342	0.94551495
130000	0.96603604	0.9380781	0.93617137	0.93468938	0.93986627	0.9391377
140000	0.96588206	0.94029318	0.95102071	0.94163124	0.94295029	0.93957001
150000	0.96579691	0.94420519	0.94797429	0.94493851	0.94814222	0.93547019

TABLE 3. Following table shows the accuracy comparison.

Total number of Samples	Accuracy Proposed Firefly + Hybrid Classifier	Accuracy Levenberg Neural	Accuracy DT	Accuracy RF	Accuracy KNN with 10 Neighbors	Accuracy Multi-SVM
20000	0.879843	0.698021	0.705784	0.572868	0.625573	0.606971
30000	0.888347	0.796345	0.792980	0.732847	0.738339	0.712649
40000	0.899885	0.799080	0.801771	0.650519	0.703690	0.663922
50000	0.86056	0.825312	0.817338	0.670058	0.734126	0.692394
60000	0.87050	0.813200	0.838516	0.678752	0.727752	0.671265
70000	0.87760	0.856572	0.847907	0.780608	0.756514	0.796163
80000	0.882863	0.855851	0.844143	0.785942	0.732397	0.760354
90000	0.907894	0.844569	0.843584	0.714551	0.648357	0.711093
100000	0.919845	0.874907	0.865439	0.769823	0.786530	0.79273
110000	0.928784	0.846907	0.875911	0.680684	0.829166	0.694803
120000	0.937846	0.854186	0.854114	0.707921	0.693158	0.707000
130000	0.942330	0.859385	0.854710	0.663074	0.665303	0.673865
140000	0.949835	0.863817	0.885310	0.713191	0.813273	0.703031
150000	0.949880	0.878202	0.8829310	0.766832	0.823818	0.735255

different network architectures, activation functions, learning rates, and tree parameters to optimize the model's performance for the specific classification task

at hand. This flexibility enables fine-tuning and customization to adapt the model to varying datasets and application requirements.

TABLE 4. Following table shows the recall comparison.

Total number of Samples	Recall Proposed Firefly + Hybrid Classifier	Recall Levenberg Neural	Recall DT	Recall RF	Recall KNN with 10 Neighbors	Recall Multi-SVM
20000	0.82851306	0.76903862	0.77835963	0.62136802	0.67039498	0.65136714
30000	0.88205147	0.8666330	0.86119612	0.79722399	0.80786026	0.77834124
40000	0.90895161	0.88138042	0.88296171	0.70745437	0.76393874	0.71859640
50000	0.92531343	0.90538539	0.90144950	0.72646733	0.79368270	0.74830512
60000	0.93597118	0.90743957	0.92066561	0.72663940	0.79531513	0.72129562
70000	0.94365591	0.93398537	0.92890995	0.85970455	0.82275806	0.87092168
80000	0.94934004	0.93747211	0.93016662	0.85347101	0.78458576	0.83555542
90000	0.95375149	0.93442573	0.93655094	0.76519746	0.70213711	0.76121151
100000	0.95725760	0.95273570	0.95003082	0.84350719	0.84656809	0.86158027
110000	0.96023499	0.94201617	0.95334988	0.73493026	0.90128402	0.74470265
120000	0.96276019	0.94922809	0.94747396	0.76382470	0.74230904	0.76098195
130000	0.96444139	0.95062452	0.94855178	0.71962663	0.71918271	0.72828543
140000	0.96644342	0.95320917	0.96099431	0.76913451	0.88762615	0.76057912
150000	0.96805712	0.96187755	0.96364734	0.83004618	0.89115407	0.80384628

TABLE 5. Following table shows the F-Measure comparison.

Total number of Samples	F-Measure Proposed Firefly + Hybrid Classifier	F-Measure Levenberg Neural	F-Measure DT	F-Measure RF	F-Measure KNN with 10 Neighbors	F-Measure Multi-SVM
20000	0.88893322	0.84172024	0.84642976	0.74311400	0.78340762	0.76944042
30000	0.91899763	0.90388979	0.90089912	0.86292217	0.86652866	0.84988948
40000	0.93602651	0.90822210	0.90888548	0.80366289	0.84192886	0.81236721
50000	0.94473215	0.92258969	0.91870045	0.81862171	0.86178616	0.83282414
60000	0.94915910	0.91885488	0.92980743	0.82169234	0.85930195	0.8166544
70000	0.95379618	0.93972397	0.93509489	0.89568708	0.87771431	0.90333092
80000	0.95433664	0.93979245	0.93479718	0.89615575	0.85916813	0.88249519
90000	0.95618139	0.93589172	0.93584496	0.84734084	0.80186387	0.84461187
100000	0.95844428	0.95110278	0.94599538	0.88895728	0.89497934	0.90164117
110000	0.96324239	0.93840495	0.94997115	0.82447107	0.92300539	0.83341241
120000	0.96129434	0.94153085	0.94252554	0.84354882	0.83290677	0.84327116
130000	0.96523806	0.94430963	0.94232091	0.81317882	0.81484704	0.82038001
140000	0.96616266	0.94670712	0.95598150	0.84668644	0.91445221	0.84065253
150000	0.96692569	0.95295945	0.95574657	0.88377394	0.91876529	0.86467787

4. **Adaptive Learning and Adaptability:** NNs are inherently adaptive and can continuously update their internal parameters through back propagation and gra-

dient descent, allowing them to adapt to changes in the data distribution over time. DTs, although static once trained, can be easily updated or retrained with

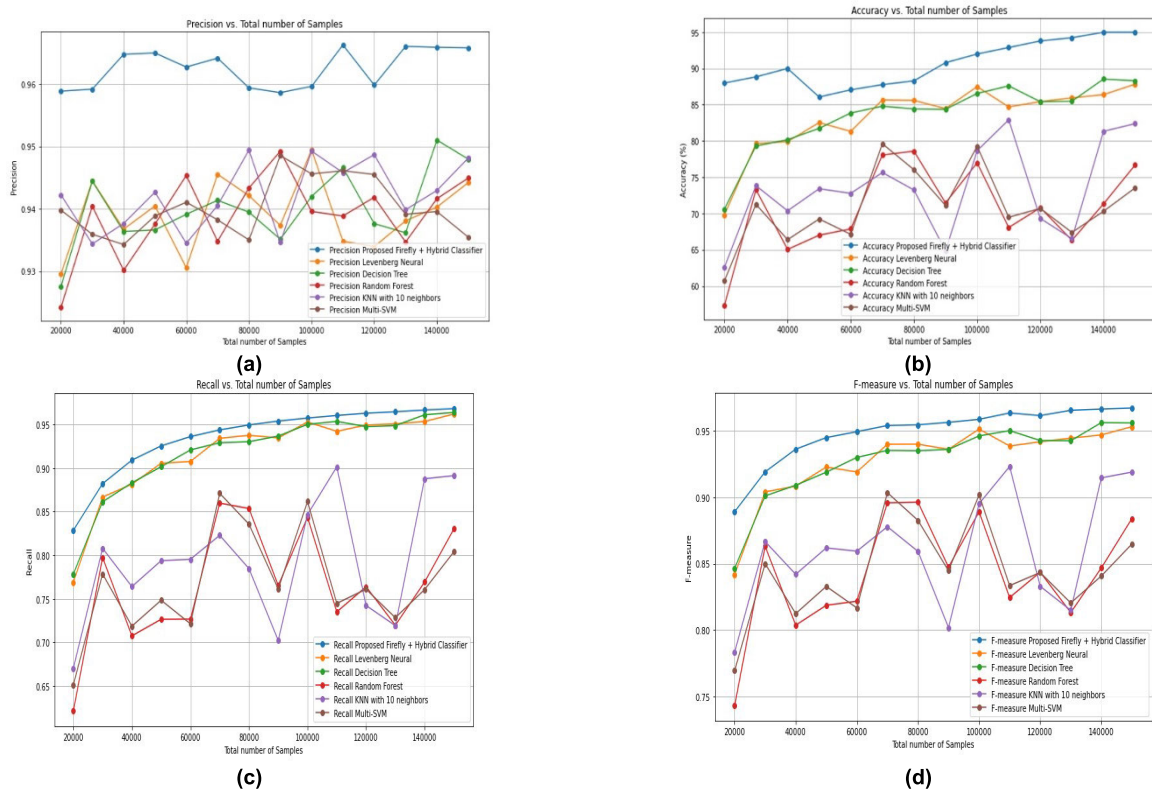


FIGURE 9. (a) Precision comparison. (b) Accuracy comparison. (c) Recall comparison. (d) F-Measure comparison.

new data to accommodate concept drift or changes in the underlying data characteristics. The hybrid model can leverage this adaptability to maintain high performance in dynamic and evolving environments, making it suitable for real-world applications where the data distribution may change over time.

5. **Enriched Model Interpretability and Explain ability:** While NN models are known for their black-box nature and lack of interpretability, DTs offer transparent and interpretable models that can provide insights into the decision-making process. By combining both models in the hybrid architecture, it not only improves classification accuracy but also enhances model interpretability and explain ability, allowing users to understand the rationale behind predictions and gain actionable insights from the model's output.

IV. PERFORMANCE EVALUATION

A. EXPERIMENT SETUP

Evaluation is one of the vital steps and this step measures the performance of the proposed methodology and compares with the existing ones. MATLAB software running on the Windows 10 operating system was utilized for this task, supported by 8 GB of RAM. The study employed both the CSE CIC IDS 2018 dataset and a simulated dataset.

In this experiment, cutting-edge machine learning evaluation metrics have been employed to assess the effectiveness of the proposed IDS. Through a review of existing literature, it's evident that Precision, Accuracy, Recall, and F-Measure stand out as the most commonly utilized evaluation metrics in machine learning.

Various test cases are created and presented the values of performance metrics in the tabular form. The evaluation of performance is measured by the Precision, Accuracy, Recall and F-Measure as in the following equations. The main performance metrics for the detection of attacks is accuracy. Accuracy is describing that how perfectly attacks are being identified by the classifier. There are some basic terms related to the performance metrics: True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN). TP refers the rate of the packets that are correctly identified as attack packets. TN value defines the rate of the normal packets identified as normal packets. FP is the rate of the attack packets identified as normal packets. It is known as Type I error. FN is the rate of the normal packets identified as attack packets. It is known as Type II error. Evaluation is one of the vital steps and this step measures the performance of the proposed methodology and compares with the existing ones. MATLAB software running on the Windows 10 operating system was utilized for this task, supported by 8 GB of RAM. The study employed both the CSE CIC IDS 2018 dataset and a simulated dataset. FFA is a nature-inspired

TABLE 6. Comparison of proposed feature selection algorithm with PSO and GA in terms of precision.

Total number of Samples	Precision PSO+Hybrid Classifier	Precision GA+Hybrid Classifier	Precision Proposed Firefly + Hybrid Classifier
20000	0.89325121	0.84606881	0.95885899
30000	0.83120870	0.81011737	0.95917421
40000	0.83774387	0.80041096	0.96476389
50000	0.89183689	0.80315651	0.96498339
60000	0.81438771	0.77887040	0.96272396
70000	0.88233581	0.91639030	0.96415675
80000	0.89576420	0.80485535	0.95938611
90000	0.88532679	0.86966180	0.95862370
100000	0.95187920	0.82428211	0.95963389
110000	0.91838369	0.77661431	0.96626869
120000	0.90554341	0.82591551	0.95983295
130000	0.81215359	0.78950563	0.96603604
140000	0.96386093	0.81487536	0.96588206
150000	0.89279464	0.84612539	0.96579691

TABLE 7. Comparison of proposed feature selection algorithm with PSO and GA in terms of accuracy.

Total number of Samples	Accuracy PSO +Hybrid Classifier	Accuracy-measure GA+Hybrid Classifier	Accuracy Proposed Firefly + Hybrid Classifier
20000	69.0787370	65.9316031	0.879843
30000	74.292018	79.4134382	0.888347
40000	74.9688163	70.2975348	0.899885
50000	75.4650278	71.3911008	0.860560
60000	82.1031559	76.395777	0.870500
70000	82.3963686	80.2633572	0.87760
80000	84.9556829	83.7484475	0.882863
90000	81.2901379	82.7116438	0.907894
100000	77.7693584	77.7328603	0.919845
110000	87.0312721	76.6258699	0.928784
120000	85.4534590	75.5824956	0.937846
130000	78.4072976	77.3573391	0.942330
140000	81.1369337	74.2547736	0.949835
150000	84.9296383	78.8100726	0.949880

optimization method that mimics the behavior of fireflies in their search for optimal lighting conditions. By applying this innovative algorithm, we aim to enhance the efficiency and accuracy of proposed intrusion detection system. Furthermore, the proposed work goes beyond mere algorithm selection. It delves into a comparative analysis by benchmarking the performance of the Firefly Algorithm against two popular optimization techniques: Particle Swarm Optimization (PSO) and Genetic Algorithms

(GA). This comparative assessment allows for a comprehensive understanding of the strengths and weaknesses of each approach, enabling the identification of the most effective method for intrusion detection within the context of this research.

In this experiment, cutting-edge machine learning evaluation metrics have been employed to assess the effectiveness of the proposed intrusion detection system. Through a review of existing literature, it's evident that

TABLE 8. Comparison of proposed feature selection algorithm with PSO and GA in terms of recall.

Total number of Samples	Recall PSO+Hybrid Classifier	Recall GA+Hybrid Classifier	Recall Proposed Firefly + Hybrid Classifier
20000	0.83396589	0.78073702	0.82851306
30000	0.88278632	0.87612737	0.88205147
40000	0.93018483	0.91105840	0.90895161
50000	0.88995665	0.82414106	0.92531343
60000	0.90343611	0.82126443	0.93597118
70000	0.84921977	0.79800564	0.94365591
80000	0.88786263	0.88940991	0.94934004
90000	0.86810371	0.85412944	0.95375149
100000	0.83271552	0.85425083	0.95725760
110000	0.80523515	0.87892040	0.96023499
120000	0.87661260	0.92861383	0.96276019
130000	0.86183583	0.76918209	0.96444139
140000	0.88628053	0.86384529	0.96644342
150000	0.86177842	0.77764284	0.96805712

TABLE 9. Comparison of proposed feature selection algorithm with PSO and GA in terms of F-Measure.

Total number of Samples	F-Measure PSO +Hybrid Classifier	F-Measure GA+Hybrid Classifier	F-Measure Proposed Firefly + Hybrid Classifier
20000	0.86902005	0.83997375	0.88893322
30000	0.82052752	0.84489217	0.91899763
40000	0.81865201	0.86043216	0.93602651
50000	0.84517686	0.84433161	0.94473215
60000	0.79623318	0.83654155	0.9491591
70000	0.89904068	0.88152732	0.95379618
80000	0.84787995	0.84432374	0.95433664
90000	0.87742438	0.86888206	0.95618139
100000	0.88349745	0.82847735	0.95844428
110000	0.84157022	0.79066581	0.96324239
120000	0.86389846	0.85050924	0.96129434
130000	0.80066949	0.82408667	0.96523806
140000	0.88312869	0.84907935	0.96616266
150000	0.86883376	0.85388018	0.96692569

Precision, Accuracy, recall, and F-Measure stand out as the most commonly utilized evaluation metrics in machine learning.

Various test cases are created and presented the values of performance metrics in the tabular form. The evaluation of

performance is measured by the Precision, Accuracy. Recall and F-Measure would be as follows defined in Eq. (2)–(5). The main performance metrics for the detection of attacks is accuracy. There are some basic terms related to the performance metrics: True Positive (TP), True Negative (TN),

TABLE 10. Following table shows the precision comparison.

Total number of Samples	Precision Proposed Firefly + Hybrid	Precision Levenberg Neural	Precision DT	Precision RF	Precision KNN with 10 Neighbors	Precision Multi-SVM
10000	0.97070109	0.94371507	0.94699337	0.95033276	0.94434593	0.94605655
20000	0.96445322	0.94656442	0.9443651	0.9475000	0.94434573	0.94210452
30000	0.96191003	0.95140165	0.94114795	0.94460563	0.94864945	0.94465708
40000	0.96054832	0.9400717	0.94519552	0.94067596	0.94077184	0.94350933
50000	0.96679479	0.94216509	0.94223591	0.94594996	0.94486697	0.93566467
60000	0.95959228	0.94733527	0.94698827	0.94761559	0.9415055	0.94730859
70000	0.9631075	0.94036768	0.94180683	0.94679845	0.94231306	0.94677086
80000	0.96332745	0.94765781	0.94137151	0.95121375	0.94557627	0.94348362
90000	0.96244337	0.93868467	0.93782452	0.94224961	0.93786857	0.94157871
100000	0.96586151	0.94163279	0.94708224	0.93809159	0.94532042	0.9408459

TABLE 11. Following table shows the accuracy comparison.

Total number of Samples	Accuracy Proposed Firefly + Hybrid	Accuracy Levenberg Neural	Accuracy DT	Accuracy RF	Accuracy KNN with 10 Neighbors	Accuracy Multi-SVM
10000	91.3200000	88.0854162	89.0024674	73.9948182	74.082397	73.6674392
20000	91.6800000	89.6750944	88.8071634	78.8322933	70.4459383	75.7203510
30000	91.7900000	90.4500109	88.7839973	80.3935722	74.550968	82.0204144
40000	91.8425000	88.9689545	89.709981	73.1233596	79.8109589	73.340178
50000	91.9100000	89.4015388	89.108862	74.2556578	73.3204334	72.2240623
60000	91.9316667	89.9367308	90.2193526	78.4939435	83.6650098	77.8595548
70000	91.9185714	88.7543378	89.3023569	72.7824099	75.2217781	73.3353638
80000	91.9325000	89.8462162	88.8295227	79.2131451	70.6162385	76.2271262
90000	91.9577778	88.8719189	88.8716406	71.6281608	70.1046478	71.2662066
100000	91.9470000	89.2660161	89.9816264	74.0890382	74.9771295	75.1947625

False Positive (FP) and False Negative (FN). TP refers the rate of the packets that are correctly identified as attack packets. TN value defines the rate of the normal packets identified as normal packets. FP is the rate of the attack packets identified as normal packets. It is known as Type I error. FN is the rate

of the normal packets identified as attack packets. It is known as Type II error.

Precision is describing that how correctly the packets are identified by the classifier. Accuracy is describing that how perfectly attacks are being identified by the classifier. Recall

TABLE 12. Following table shows the recall comparison.

Total number of Samples	Recall Proposed Firefly + Hybrid	Recall Levenberg Neural	Recall DT	Recall RF	Recall KNN with 10 Neighbors	Recall Multi-SVM
10000	0.97877814	0.97084782	0.97377555	0.79601528	0.80350511	0.79427382
20000	0.98416617	0.9810388	0.97685381	0.85244272	0.76020434	0.82583194
30000	0.98589381	0.98310249	0.98174377	0.87725407	0.8010276	0.89093119
40000	0.98667848	0.98207006	0.9836016	0.79384528	0.87129868	0.7938394
50000	0.98762116	0.98426527	0.98424038	0.80082892	0.79007506	0.78572233
60000	0.98801677	0.98456373	0.98579326	0.84812138	0.91559615	0.84236008
70000	0.98790131	0.98357569	0.98402106	0.78342863	0.81594429	0.78859459
80000	0.98812307	0.98491968	0.98269374	0.85199383	0.75993718	0.82848737
90000	0.98843903	0.98356653	0.98487059	0.77380211	0.76094323	0.770193
100000	0.98839047	0.98524936	0.98487797	0.80814008	0.81134415	0.81796635

TABLE 13. Following table shows the F-Measure comparison.

Total number of Samples	F-Measure Proposed Firefly + Hybrid	F-Measure Levenberg Neural	F-Measure DT	F-Measure RF	F-Measure KNN With 10 Neighbors	F-Measure Multi-SVM'
10000	0.97472288	0.95708918	0.96019774	0.86635582	0.86825108	0.86354632
20000	0.97420999	0.96349333	0.96033475	0.89746131	0.8423287	0.88014476
30000	0.97375426	0.96699233	0.96101733	0.90968491	0.86861103	0.91700789
40000	0.97343808	0.96061205	0.96401619	0.86104586	0.90470349	0.86222745
50000	0.97709701	0.96275515	0.96278022	0.86736117	0.86056573	0.8541631
60000	0.9735971	0.9655908	0.96600121	0.89511221	0.92837009	0.89175717
70000	0.97534686	0.9614865	0.96245127	0.85740076	0.87458752	0.86047395
80000	0.97556773	0.96592952	0.9615889	0.89887404	0.84265365	0.88225401
90000	0.9752680	0.96060164	0.96077197	0.84975846	0.84019282	0.84730613
100000	0.97699613	0.96294742	0.9656104	0.86828045	0.87322328	0.87511363

measure is similar to the detection rate measure. F-Measure is identified as the harmonic mean of the precision and recall measure. Eq. (2)–(5) are shown in the following section of the paper. Performance metrics are described in equation forms as Eq. (2)–(5).

The proposed work presented in this study represents a significant step forward in the field of intrusion detection. Leveraging a substantial dataset comprising 1.5 lakh (150,000) records, this research focuses on the identification and classification of network-based

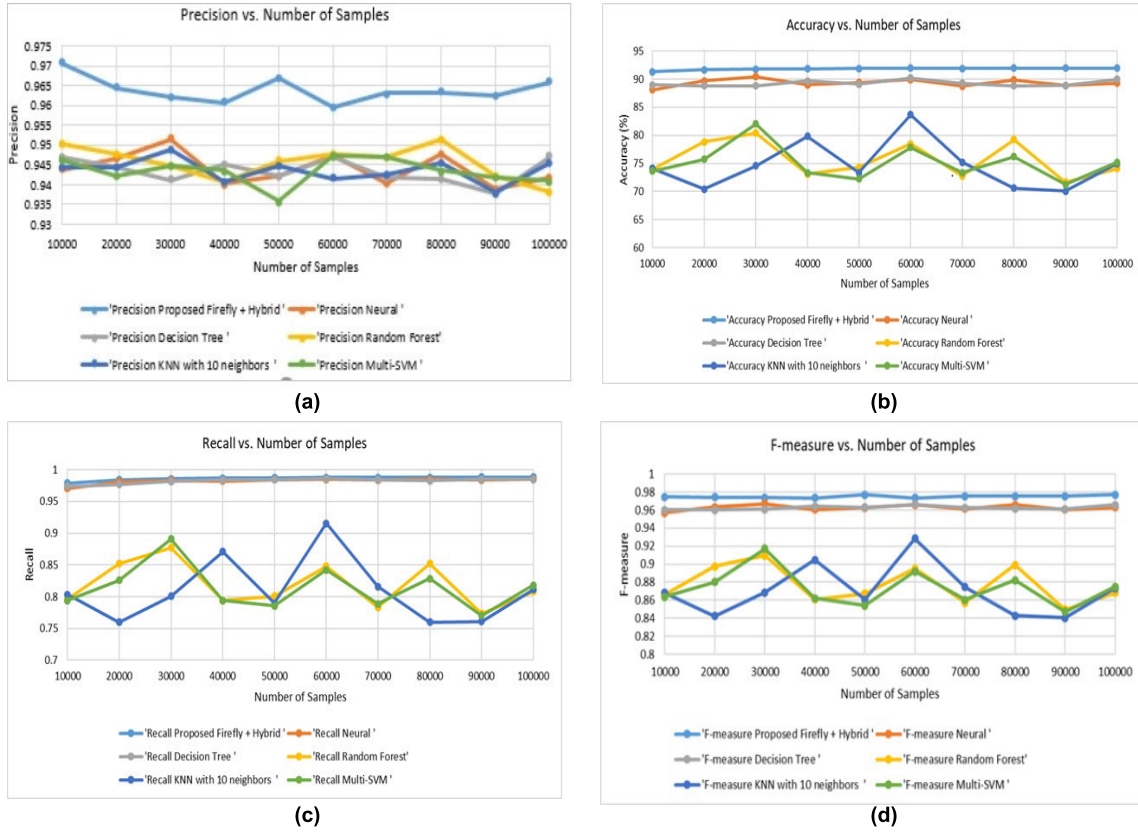


FIGURE 10. (a) Precision comparison. (b) Accuracy comparison. (c) Recall comparison. (d) F-Measure comparison.

attacks. Within this extensive dataset, five major categories of attacks have been meticulously categorized, serving as representative examples of common cyber threats.

The fundamental objective of this research is to develop an intrusion detection system that not only identifies attacks but also minimizes false positives, thus optimizing the precision of detection. This is achieved through the comprehensive evaluation of detection performance metrics, including Precision, Accuracy, Recall, and F-Measure. These metrics collectively offer a holistic view of the system's efficacy, capturing both its ability to accurately identify attacks and its capacity to minimize false alarms. What sets this research apart is its utilization of a novel computational intelligence technique known as the FFA.

The 'Total number of Samples' column represents the dataset sizes, ranging from 20,000 to 150,000 samples, simulating diverse network scenarios. Notably, the 'Precision Proposed Firefly + Hybrid Classifier' consistently stands out, achieving high precision scores exceeding 0.95 across all dataset sizes. This demonstrates its remarkable ability to correctly identify attacks while minimizing false positives. In contrast, other classifiers, such as Levenberg Neural, DT, RF, K-nearest Neighbor (KNN) with 10 Neighbors, and Multi-SVM, exhibit varying precision scores with changing dataset sizes. Moreover, the proposed Firefly Algorithm + Hybrid Classifier maintains its competitive edge in precision, even as the dataset size increases, underscoring its robustness and suitability for real-world intrusion detection scenarios. This data reinforces the algorithm's effectiveness

$$\text{Precision} = \frac{\text{True Positive}}{(\text{True Positive} + \text{False Positive})} \quad (2)$$

$$\text{Accuracy} = \frac{(\text{True Positive} + \text{True Negative})}{(\text{True Positive} + \text{True Negative} + \text{False Positive} + \text{False Negative})} \quad (3)$$

$$\text{Recall} = \frac{\text{True Positive}}{(\text{True Positive} + \text{False Negative})} \quad (4)$$

$$\text{F-measure} = \frac{(2 * \text{Precision} * \text{Recall})}{(\text{Precision} + \text{Recall})} \quad (5)$$

and reliability, making it a compelling choice for enhancing cybersecurity in networks of varying scales.

Notably, the 'Accuracy Proposed Firefly + Hybrid Classifier' consistently stands out, achieving high Accuracy scores exceeding 0.95 regardless of the data size. This demonstrates its remarkable ability to correctly identify attacks while minimizing false positives. The proposed algorithm performs better in terms of accuracy even when dealing with large or complex datasets.

The presented table offers a valuable glimpse into the recall performance of multiple classifiers across a spectrum of dataset sizes, ranging from 20,000 to 150,000 samples, mirroring diverse network scenarios. Notably, the 'Recall Proposed Firefly + Hybrid Classifier' consistently emerges as the top-performing classifier, achieving impressive recall scores that steadily ascend from 0.8285 at 20,000 samples to an exemplary 0.9681 at 150,000 samples. This signifies the algorithm's remarkable effectiveness in identifying true positive instances, a pivotal aspect of intrusion detection. The 'Recall Levenberg Neural' classifier also demonstrates competitive performance, gradually improving from 0.7690 to 0.9619 across the dataset sizes. The 'Recall DT' maintains consistent recall scores, indicating its proficiency in attack detection, especially in larger-scale networks. While the 'Recall RF' exhibits competitive performance, it falls slightly behind the top-performing classifiers. Similarly, the 'Recall KNN with 10 Neighbors' and 'Recall Multi-SVM' classifiers prove their effectiveness, with the former demonstrating significant improvement as the dataset size increases. In sum, this data underscores the robustness and reliability of the 'Recall Proposed Firefly + Hybrid Classifier' for intrusion detection, positioning it as a compelling choice for network security where accurate attack identification is paramount.

Notably, the 'F-Measure Proposed Firefly + Hybrid Classifier' consistently emerges as a top-performing classifier, achieving F-Measure scores that steadily increase from 0.8889 at 20,000 samples to a remarkable 0.9669 at 150,000 samples. This signifies the algorithm's proficiency in striking a balance between precision and recall, a vital aspect of intrusion detection. The 'F-Measure Levenberg Neural' classifier also demonstrates competitive performance, gradually improving from 0.8417 to 0.9529 across the dataset sizes. The 'F-Measure DT' maintains consistent scores, indicating its robustness in attack detection, particularly in larger-scale networks.

While the 'F-Measure RF' exhibits competitive performance, it lags slightly behind the top-performing classifiers. Similarly, the 'F-Measure KNN with 10 Neighbors' and 'F-Measure Multi-SVM' classifiers prove their effectiveness, with the former showcasing significant improvement as the dataset size increases. In conclusion, this data underscores the robustness and reliability of the 'F-Measure Proposed Firefly + Hybrid Classifier' for intrusion detection, making it an enticing choice for network security where achieving a balanced performance between precision and recall is crucial. The 'F-Measure Levenberg

Neural' classifier also emerges as a strong contender, highlighting its suitability for this task.

Simulated dataset using the cloudsim toolkit results in the following results which are presented in the tabular and graphical forms.

Precision values vary with sample size, indicating that model performance may be influenced by the amount of data available for training and testing. The 'Precision Proposed Firefly + Hybrid' model consistently has the highest precision across different sample sizes, making it a strong performer in this dataset. Its important to consider not only precision but also other evaluation metrics and practical considerations when choosing a machine learning model for a specific task.

The 'Accuracy Proposed Firefly + Hybrid' model consistently has high accuracy across different sample sizes, making it a strong performer in terms of overall correctness. The 'Accuracy RF', 'Accuracy KNN with 10 Neighbors', and 'Accuracy Multi-SVM' models show the most significant variability in performance, with lower accuracy values observed for some sample sizes.

The 'Accuracy Neural' and 'Accuracy DT' models perform competitively, with good accuracy values, although they may exhibit some variability.

The 'Recall Proposed Firefly + Hybrid', 'Recall Neural', and 'Recall DT' models generally exhibit high recall values, making them strong performers in terms of correctly identifying relevant instances. The Recall RF model shows the most variability in performance across different sample sizes, with lower recall values observed for some cases. The 'Recall KNN with 10 Neighbors' and 'Recall Multi-SVM' models exhibit variability in recall values, indicating that their performance may be influenced by the specific dataset or sample size.

The 'F-Measure Proposed Firefly + Hybrid', 'F-Measure Neural', and 'F-Measure DT' models consistently exhibit high F-measure values, indicating their effectiveness in achieving a balance between precision and recall. The 'F-Measure RF' model shows the most variability in performance, with lower F-Measure values for some sample sizes. The 'F-Measure KNN with 10 Neighbors' and 'F-Measure Multi-SVM' models also demonstrate variability in F-Measure values, suggesting that their performance may be influenced by the specific dataset or sample size.

V. CONCLUSION AND FUTURE DIRECTIONS

A unique approach for creating an intrusion detection system was presented in this study. The method involves combining the hybrid firefly algorithm with the hybrid classifier. The recent CSE CIC IDS 2018 dataset and simulated dataset were used to assess the effectiveness of the proposed architecture. Novel feature selection algorithm is proposed which is the combination of the firefly algorithm with the decision tree. The proposed feature selection performs better than the PSO and GA. We have studied in the literature review that FFA is performing better than PSO and GA. Also we observed

this by performing practically that results are better with the proposed feature selection algorithm. Hybrid classifier is used which is the hybridization of neural network with the DT. The proposed architecture outperforms the other state-of-the-art techniques for finding attacks in the cloud computing environment.

The future directions related to proposed work is itemized below:

- Adaptive attack detection system is a good future scope in the field of security of clouds. Dynamic conditions can be controlled by developing an adaptive detection system. Dynamic network condition include change in the environmental configurations, computation resources and different locations where attack detection systems are deployed. Dynamic conditions can be controlled by developing an adaptive detection system.
- Another future direction can be developing an IDS which expands or contracts according to the virtual machines of the cloud. Vulnerabilities can be detected by discovering an efficient detection system.

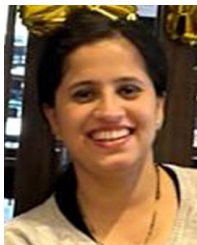
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IN-HO RA (Member, IEEE) received the Ph.D. degree in computer engineering from Chung-Ang University, Seoul, South Korea. He is currently a Professor. His research interests include wireless ad hoc and sensor networks, blockchain, the IoT, PS-LTE, and microgrids.



POOJA RANA received the B.Tech. and M.Tech. degrees in computer science and engineering from Punjab Technical University, Jalandhar. She was an Assistant Professor. She is currently a Research Scholar with Lovely Professional University, Phagwara. Her research interests include cloud computing and security.



ISHA BATRA received the M.E. and Ph.D. degrees. She is currently an Associate Professor with Lovely Professional University, India. She has over ten years of teaching experience and has published around 30 research papers and articles. She has supervised nine M.Tech. and supervising eight Ph.D. dissertations. Her research interests include wireless networks, sensor networks, ad hoc networks, VANET, MANET, and the Internet of Things.



ARUN MALIK received the M.Tech. and Ph.D. degrees. He is currently a Professor with Lovely Professional University, India. He has more than 13 years of teaching experience and has published around 60 research papers in SCI/Scopus-indexed journals and conferences. He has supervised eight M.Tech. students and supervising eight Ph.D. dissertations. His current research interests include wireless networks, ad hoc networks (VANET and MANET), the Internet of Things (IoT), and network security.



OH-SUNG LEE received the bachelor's degree in electronic and electrical engineering from Hanyang University, South Korea, in 2008, and the master's degree in business administration from Chonbuk National University, in 2018. He is currently pursuing the Ph.D. degree in information and communications propagation engineering with Kunsan National University. His research interests include cloud, blockchain, and the Internet of Things.



A. S. M. SANWAR HOSEN received the M.S. and Ph.D. degrees in computer science and engineering from Jeonbuk National University, Jeonju, South Korea. He is currently an Assistant Professor with the Department of Artificial Intelligence and Big Data, Woosong University, Daejeon, South Korea. He has published several papers in journals and international conferences. His research interests include wireless sensor networks, the Internet of Things, fog cloud computing, cyber security, data distribution services, artificial intelligence, blockchain, and green IT. He has been an Expert Reviewer of IEEE TRANSACTIONS, Elsevier, Springer, and MDPI journals and magazines. He has also been invited to serve as the Guest Editor and the Program Committee Member for several reputed international conferences, such as IEEE and ACM.

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